

requirements describe how to look at the performance of the platform you have chosen. The OS performance parameters, reliability and availability aspects under the 'X-ability' section are discussed as separate articles. Here, we focus on what the software performance parameters are. The entire set of non-functional requirements, and all the different parameters under each of these categories, are shown in Figures 1 through 7.

The quality of product architecture

How do we judge whether an architecture is good or not so good? Is it possible to quantify architecture? There are many parameters to measure architecture quality; some of the important ones are listed here:

1. Efficiency or performance
2. Reliability and availability
3. Modifiability / maintainability / scalability
4. Security
5. Interoperability and portability

The efficiency of a product would mean the data rate, throughput or error rate, and other performance parameters, based on the product. Reliability was briefly touched upon in the previous article (*LFY* June 2010 issue). Availability is related to reliability, and they come under the category of 'X-ability' parameters. Availability is measured in terms of three 9s (99.9% available), four 9s (99.99% available) or five 9s (99.999% available).

Security is also an important parameter in measuring how good an architecture is. The amount of security we provide in the product can be quantified in terms of three or more levels of logging in and password protection, plus 128-bit key encryption, which is a standard nowadays.

Now let's suppose that we have an application running on a remote server on a remote network, and we are connected to a local network. We have to log in to the remote network first, then log in to the remote server, and finally, log on to the application. Here, there are three levels of security. Another example is an application that does financial transactions, and is in a PCI-compliant environment. To interact with that application, we need to go through multiple levels of authentication, such as:

1. An RSA token is needed for user authentication.
2. Next, you log in to the server.
3. Finally, you log in to the application.

Interoperability is the ability of a product to work with other vendors' products. For example, I have a Sony camcorder. Can it work with a Panasonic DVD? Can a Panasonic MP3 player play MP3 and MP4 also? If your product can work with at least three other vendors' products, it is commendable.

Portability is the ability of software to work on multiple platforms. Let's take a product in which an application 'A' is written in C on VxWorks. Can we port this application easily to Linux, with minimum changes to the application? If so,

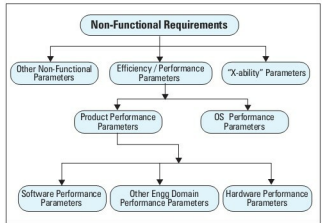


Figure 1: Non-functional requirements

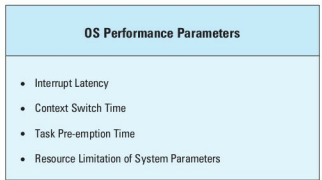


Figure 2: OS performance parameters

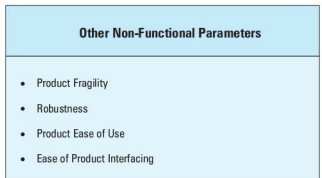


Figure 3: Other non-functional parameters

the product is portable. Another way to look at portability is the ability of the product to work on multiple hardware architectures. If a product is developed on the ARM platform, can it work on the Intel x86 platform too?

These five parameters are some recommendations (or preferences) of what can be used to measure the quality of architecture. This has the backing of theory too, in Bass, Clements and Kazman's 'Software Architecture in Practice'. However, more details are needed when we talk about how to measure these parameters.

All these requirements are at the product level, not just confined to software or hardware. The actual parameters